

Alex Shuey

917-826-0620 - ans242@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science in Mechanical Engineering

May 2027

PROFESSIONAL EXPERIENCE

Cornell University Sibley School of Mechanical Engineering
Student Researcher

May 2025-Present

- Worked with a 3-person student team under Mason Peck to develop the first production Magnus Effect-powered ultralight aircraft.
- Utilized SOLIDWORKS, ANSYS, and MATLAB to create a design capable of consistent performance for the collection of scientific data.
- Fabricated and flight-tested VTOL prototype, conducting ten five-minute hovers while lifting 4 kg of external payload (25% above initial goal).
- Designed and fabricated a dynamically balanced, gear-driven hub system for 1,500+ RPM Magnus Effect sphere actuation.

Cornell University Office of Visitor Relations
Information Specialist

May 2024 – Present

- Serve as an information specialist, providing excellent customer service to all university visitors.
- Conduct campus tours, manage the front desk, and answer the Cornell University switchboard.

Cornell University Design Build Fly
Propulsion Sub-Team Lead

November 2023 – Present

- Lead a team of 6 undergraduate students
- Delegate tasks and responsibilities for designing the propulsion system of a radio-controlled aircraft

Propulsion Sub-Team Member

- Identify the battery and motor specifications required for the necessary flight time and payload capacity.
- Perform propeller static thrust tests, ensuring the achievement of targeted metrics.
- Design, test, integrate, iterate, solder, and wire the plane's propulsion system.
- Collaborate with aerodynamics and mechanical sub-teams to oversee the design and fabrication of radio-controlled aircraft.

LEADERSHIP & COMMUNITY INVOLVEMENT

Society of Women Engineers
Outreach Committee Coordinator

September 2023 – Present

- Organize weekly meetings to create middle school outreach events to interest students in STEM.
- Lead a committee of 7 colleagues.

Cornell University College of Engineering
Engineering Ambassador

September 2023 – Present

- Serve as a representative of the College of Engineering, leading virtual and in-person sessions to engage and guide prospective students.
- Participate in outreach initiatives, including virtual panels and in-person events.

SKILLS & INTERESTS

Skills: CAD (Solidworks, Fusion 360), MATLAB, Soldering, Ecalc, Machining (Mill, Lathe), Arduino, Customer Service, Public Speaking, Aviation Trainee, Teamwork, Tutoring

Interests: Flying/Aviation, Space Exploration, Electric Aircraft Development, Sustainability